The empirical recurrence for OEIS A220914, proved by transfer matrix

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Abstract

OEIS A220914 counts the distinct indicator arrays obtained from the $n \times 3$ arrays over $\{0, \ldots, 2\}$ by marking the cells whose value is equal to exactly one of their neighbours, and carries an empirical recurrence of order 63 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. What is counted is the IMAGE of a map, not its domain, so it is not a walk count in the obvious graph; the determinisation of the machine that emits the indicator row by row turns it into one, on $S = 445$ states, after which the conjectured recurrence is settled by a finite exact computation.

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1 The conjecture

OEIS A220914 is “Equals one maps: number of nX3 binary arrays indicating the locations of corresponding elements equal to exactly one of their horizontal, diagonal and antidiagonal neighbors in a random 0..2 nX3 array.”. It has offset 1 and begins

$$4, 37, 379, 3299, 27365, 220299, 1754781, 13920341, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 11*a(n-1) -13*a(n-2) -88*a(n-3) -179*a(n-4) +963*a(n-5) +3374*a(n-6) -4092*a(n-7) -27752*a(n-8) -54928*a(n-9) +159273*a(n-10) +565084*a(n-11) +262024*a(n-12) -2548728*a(n-13) -9489404*a(n-14) -3862504*a(n-15) +48520232*a(n-16) +136801104*a(n-17) +34658552*a(n-18) -677179488*a(n-19) -1711015536*a(n-20) +28469504*a(n-21) +7647579792*a(n-22) +17156152832*a(n-23) +10757172640*a(n-24) -51483411712*a(n-25) -179983506432*a(n-26) -190932391040*a(n-27) +213395380864*a(n-28) +783959354112*a(n-29) +1174739137792*a(n-30) +749321041920*a(n-31) -2051932146688*a(n-32) -5471137934336*a(n-33) -4763822426112*a(n-34) -1136691800064*a(n-35) -1642622783488*a(n-36) -10474931142656*a(n-37) -3637079998464*a(n-38) +36568785813504*a(n-39) +99840437256192*a(n-40) +144085753987072*a(n-41) +174717893804032*a(n-42) +188630020718592*a(n-43) +93143850483712*a(n-44) -49825152630784*a(n-45) +17014351462400*a(n-46) +14859929911296*a(n-47) -10286599241728*a(n-48) +133670360317952*a(n-49) -132373305884672*a(n-50) -213409776795648*a(n-51) +27968935034880*a(n-52) -114439382630400*a(n-53) -53710499610624*a(n-54) +22803071893504*a(n-55) -36764240576512*a(n-56) -26657024901120*a(n-57) -3002651901952*a(n-58) +859630993408*a(n-59) -146968412160*a(n-60) -5053029023744*a(n-61) +3498250862592*a(n-62) -1013612281856*a(n-63)

As of the “Last modified” line on the live entry (Oct 03 2025 19:06:19, revision 9) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

Let $R = \{0, \dots, 2\}^3$ be the set of rows and let the neighbours of a cell be the cells at the offsets

$$(-1, -1), (-1, 1), (0, -1), (0, 1), (1, -1), (1, 1),$$

those falling outside the array being simply absent. For an array $A = (r_1, \dots, r_n) \in R^n$ let $\Phi(A) \in \{0, 1\}^{n \times 3}$ mark the cells whose value is equal to exactly one of its neighbours. The entry counts

$$a(n) \;=\; \bigl|\{\Phi(A) : A \in R^n\}\bigr|,$$

the size of an IMAGE. This is the whole difficulty: two different arrays with the same marking must be counted once, and a walk count in a graph on the arrays would count them twice.

Because every offset above moves the row index by at most one, the marked row i depends only on rows $i-1, i, i+1$. Write $\beta(p, c, x) \in \{0, 1\}^3$ for the marking of the row c with p above and x below, allowing p or x to be the symbol $\perp$ for "outside". Then

$$\Phi(r_1, \dots, r_n) = \bigl(\beta(\perp, r_1, r_2),\ \beta(r_1, r_2, r_3),\ \dots,\ \beta(r_{n-1}, r_n, \perp)\bigr).$$

3 The image is a walk count after determinisation

For a set $D \subseteq (R \cup \{\perp\}) \times R$ of pairs and a row $b \in \{0, 1\}^3$ put

$$\delta(D, b) = \{(c, x) \;:\; (p, c) \in D,\ x \in R,\ \beta(p, c, x) = b\}, \qquad f(D) = \bigl|\{\beta(p, c, \perp) : (p, c) \in D\}\bigr|,$$

and let $D_0 = \{(\perp, r) : r \in R\}$.

Lemma 1. *For every $t \geq 0$ and every $b_1, \dots, b_t$, the set $D_t = \delta(\dots\delta(D_0, b_1)\dots, b_t)$ is exactly*

$$\{(r_t, r_{t+1}) \;:\; (r_1, \dots, r_{t+1}) \in R^{t+1},\ \beta(\perp, r_1, r_2) = b_1,\ \dots,\ \beta(r_{t-1}, r_t, r_{t+1}) = b_t\},$$

with $r_0 = \perp$. Consequently, for $n \geq 1$,

$$a(n) \;=\; \sum_{b_1, \dots, b_{n-1}} f(D_{n-1}),$$

the sum over all sequences of rows in $\{0, 1\}^3$, empty D's contributing nothing.

Proof. Induction on t. For $t = 0$ the set is $\{(\perp, r_1)\}$, which is D_0. If the description holds for D_t, then $\delta(D_t, b)$ collects the pairs (r_{t+1}, r_{t+2}) for which (r_t, r_{t+1}) already extends a consistent prefix and $\beta(r_t, r_{t+1}, r_{t+2}) = b$, which is the description for $t + 1$.

An element of the image is a full marking $(b_1, \dots, b_n)$. By the description, a prefix $(b_1, \dots, b_{n-1})$ is realised by some $r_1, \dots, r_n$ exactly when $D_{n-1} \neq \emptyset$, and then the possible last rows are precisely $\{\beta(p, c, \perp) : (p, c) \in D_{n-1}\}$, of which there are $f(D_{n-1})$. Distinct sequences $(b_1, \dots, b_n)$ are distinct markings, so summing f over the prefixes counts the image exactly once each. $\square$

So with M the adjacency matrix of the digraph whose vertices are the reachable nonempty sets D and which carries one edge $D \to \delta(D, b)$ for each b with $\delta(D, b) \neq \emptyset$ — several b may lead to the same set, and then the edge is repeated, which is exactly right, since the markings differ —

$$a(n) \;=\; \iota^\top M^{n-1} \tau, \qquad \iota = e_{D_0}, \quad \tau = f.$$

There are $S = 445$ reachable sets, generated from D_0 outward, and a is C-finite. Note that τ is not the all-ones vector: the last marked row is not a step of the walk but a count attached to the state the walk stops in.

4 The proof

Lemma 2. *Let $q(t) = t^{63} - \sum_i c_i t^{63-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+63} - \sum_i c_i M^{j+63-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 3.

$$a(n) \;=\; 11\,a(n-1)-13\,a(n-2)-88\,a(n-3)-179\,a(n-4)+963\,a(n-5)+3374\,a(n-6)-4092\,a(n-7)-27752\,a(n-$$

for every $n > 63$.

Proof. The vector $w = q(M)\tau$ was formed by 63 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 63 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the machine is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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